

SUNDAY ASPITA ABRAHAM

Machine Learning & NLP Researcher

Amfani House, Ibadan South West, Oyo, Nigeria 200252 | +234 813 148 2071 | godwinsunday11223@gmail.com | [LinkedIn](#) | [GitHub](#) | [Hugging Face](#) | [Portfolio](#)

RESEARCH INTERESTS

Low-resource NLP · African language technology · multilingual evaluation · generative AI safety · document intelligence

EDUCATION

Bowen University, Iwo, Osun State — B.Sc. Computer Science (First-Class Honours)
Nov 2020 – Nov 2024 - CGPA 4.86/5.00; 2nd Best Student; Matric No. BU21CSC2006.
Carnegie Mellon University Africa, Kigali, Rwanda — M.S. Information Technology (MSIT)
Admitted Fall 2026; declined due to lack of funding.

RESEARCH EXPERIENCE

Fatima Fellowship — Research Intern, Document Intelligence
Feb 2026 – Present - Built a public Gradio leaderboard for the ADIP Arabic Document AI Benchmark on [Hugging Face Spaces](#), supporting reproducible evaluation of Arabic OCR and document-understanding models. - Contributed to Arabic/African-language OCR and PDF-parsing research; prototyped multilingual QA and low-resource OCR pipelines. - Co-authoring the ADIP benchmark paper, submitted to the **Arabic NLP Workshop @ ACL 2026**.

Bowen University — Undergraduate Researcher, AI & Computer Vision
Jan 2024 – July 2024 - Plant-disease detection from 1,742 field photographs: MobileNetV2 transfer learning achieved **97.96% tomato** and **98.39% cassava** accuracy. - Graph Spectral Entropy metric for evaluating AI-generated industrial safety instructions; selected for the **IEEE IES Generative AI Challenge 2026**.

NYSC Data Analysis Group, Iseyin Camp — Data Analyst
Jan 2025 – Feb 2025 - Analysed 1,500+ corps-member survey responses; identified welfare and administration gaps and presented recommendations adopted by camp leadership.

SELECTED PROJECTS

- **ProverbEval-Yoruba** — Cultural-reasoning benchmark with 3,000+ Yoruba proverbs; measures whether LLMs understand culturally embedded meaning or rely on Westernized shortcuts across GPT-4o, Gemini, and DeepSeek.
- **Multi-API Disagreement for African MT** — Reference-free quality estimation for English→Yoruba/Swahili/Hausa on FLORES-200; strong-consensus filtering reaches **AUC 0.642**, with label-sensitivity up to **0.885** for Yoruba.
- **AfriKnow** — 350-question benchmark exposing calibration disparity between African and European knowledge in LLMs; Llama models are significantly overconfident on wrong answers about Africa (**Cohen's d = 1.00, p < .0001**).

RESEARCH & PUBLICATIONS

- **Bridging the Domain Gap: Transfer Learning and Aggressive Fine-Tuning for Robust Plant Disease Detection in Low-Resource African Agriculture** — *PMLR 319 (2026)*. [\[paper\]](#)
- **Graph Spectral Entropy: A Neuro-Symbolic Quality Control Metric for Industrial Generative AI Safety Instruction Verification** — *IEEE IES Generative AI Challenge 2026*.
- **AfriKnow: Calibration Disparity on Culturally Sensitive African versus European Knowledge in LLMs from Two Families** — *under review*.
- **ProverbEval-Yoruba: Cultural Reasoning Evaluation** — *in preparation for EACL 2027*.
- **ADIP: Open Arabic Document AI Benchmark** — *co-author; submitted to Arabic NLP Workshop @ ACL 2026*.

POSTERS & PRESENTATIONS

- **Harnessing Artificial Intelligence for Public Health Insights: A Machine Learning Approach to Vaccine Adoption Patterns in Nigeria** — *Poster, AI in Health Africa Conference, Makerere University*. [\[poster\]](#)

TECHNICAL SKILLS

Languages & tools: Python, SQL, Git, LaTeX, Excel, Power BI, Google Sheets, Streamlit, Hugging Face, Gradio.
ML & AI: PyTorch, TensorFlow, scikit-learn, Transformers, MobileNetV2, transfer learning, LLM evaluation, prompt engineering, OCR pipelines, spaCy.
Domains: Low-resource NLP, document intelligence, African language technology, AI safety evaluation, computer vision, survey analysis.

CERTIFICATIONS

DataCamp Associate Data Engineer / Data Analyst / Data Scientist / AI Fundamentals / Data Literacy; IBM SkillsBuild Python/Data Analysis/Data Visualization; HIIT Certified Data Analyst; UN Women / NO MORE Global Summit on Tech-Facilitated Gender-Based Violence (2026).

LEADERSHIP & COMMUNITY

- **Zindi Country Ambassador (Nigeria)** — mentored 50+ students in data-science competitions.

- **Technical Reviewer, MENA ML Winter School 2026 (KAUST)** — reviewed 2,400+ applications.
- **Virtual Judge, Deep Learning Indaba 2025** — evaluated 50+ African AI projects.
- **Machine Learning Collective (MLC)** — Research Affiliate; co-authored the PMLR 2026 plant-disease paper.
- **Data Science Nigeria (DSN) Bootcamp** — alumnus/participant; **Mentor, Wale Lab** — recognised March 2026.

LANGUAGES

English (fluent), Yoruba (native), Hausa (basic greetings).